Jianchao Wang

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Education

Nanjing University of Aeronautics and Astronautics 2019.09–2023.06
B.Eng. in Software Engineering (GPA: 4.0 / 5.0) *Nanjing, China*
Nanjing University of Aeronautics and Astronautics 2023.09–2026.04
M.Eng. in Computer Science (GPA: 4.4 / 5.0) *Nanjing, China*

Research

3D Gaussian Splatting Optimization & Artifact Removal 2024.09–2026.05
Low-Frequency First: Eliminating Floating Artifacts in 3DGS *Best Paper Honorable Mention @ CW2025*

- **Problem:** Identified floating artifacts as a key limitation in 3DGS-based editing/reconstruction tasks.
- **Method:** Proposed a frequency-domain analysis framework to pinpoint artifact causes; developed EFA-GS, a lightweight optimization strategy requiring no additional computational overhead.
- **Result:** Effectively eliminated artifacts across multiple 3DGS variants; code open-sourced (**20+ GitHub stars**); 2 citations on arXiv since Aug 2025.
- **Status:** Submitted to ACM MM 2025 (Score: 5.75/10); awarded Best Paper Honorable Mention @ CyberWorlds 2025; extended version accepted by *The Visual Computer* (Journal, recommended by CW committee).

Diffusion Models for Generalized Zero-Shot Learning 2023.09–2024.08
Frequency-Aware Diffusion for GZSL *Under Review @ Machine Intelligence Research*

- **Problem:** Diffusion models historically underperform on Generalized Zero-Shot Learning (GZSL) due to inherent frequency bias; limited prior work in this direction.
- **Method:** Analyzed frequency-domain behavior of diffusion sampling; proposed targeted improvements to both sampling algorithm and network architecture.
- **Result:** Achieved **10% performance gain** on standard GZSL benchmarks.
- **Status:** Currently under major revision at *Machine Intelligence Research*; Chinese invention patent filed (No. 202512020541.1).

Projects

Classifier-Free Guidance Diffusion Implementation 2022.12–2023.06
PyTorch reproduction of classifier-free sampling *200+ GitHub stars*

- Reproduced the classifier-free guidance method from scratch; Code adopted by 5+ subsequent projects.
- Repository: `https://github.com/jcwang-gh/classifier-free-diffusion-guidance-Pytorch`

Label-Noise-Robust Data Augmentation for Multi-Label Detection 2022.05
Chinese Invention Patent Granted *No. ZL202210149500.6*

- Developed a data augmentation technique robust to label noise; patent granted in 2022.

Technical Skills

Languages: Python, C/C++, Shell Script, LaTeX
Frameworks: PyTorch, Git, Linux (Ubuntu/Arch), Docker
Research Areas: Feed-forward Optimization of 3D Reconstruction, Pose-free 3D Reconstruction, VGGT, 3D Gaussian Splatting, NeRF, Diffusion Models, Generative Models, etc.
Classical ML: SVM, Random Forest, AdaBoost, K-means
Engineering: Multi-GPU training
English: CET-6: 583; TOEFL: 84(2019)

Achievements

- Open Source Contribution Award, NUAA & OpenAtom, 2025
- Best Paper Honorable Mention, CW, 2025
- Second Prize, Lanqiao Cup National Software and Information Technology Professionals Competition (Jiangsu Provincial Division, C++ Category – Group A), 2021 & 2022
- Multiple Undergraduate/Graduate Academic Scholarships (First to Third Class), NUAA, 2019–2023